Homework 3.2 – Submission Report (AI / Cloud / ML)

1. Alarm data in CSV (AI tool)

Delivered in this repository:

- `data/alarms.csv` — 2,000 rows of synthetic alarm records with columns: `id`, `duration_sec`, `peak_amplitude`, `repetition_count`, `hour_of_day`, `is_false_alarm`.
- `scripts/AI_PROMPT_FOR_ALARMS.md` — Ready-to-paste **AI tool prompt** to regenerate or extend the dataset with explicit column definitions and realism rules.
- `scripts/generate_alarms_csv.py` — **Reproducible** generator (fixed RNG seed) used to create the committed CSV when a fully offline pipeline is preferred.

Narrative: False alarms (`is_false_alarm = 1`) skew toward shorter duration, lower peak amplitude, and higher repetition count than true alarms, with intentional overlap so learning is non-trivial.

homework-3.2 > data > alarms.csv

```csv
id,duration_sec,peak_amplitude,repetition_count,hour_of_day,is_false_alarm
1,237.31,0.4918,4,7,0
2,178.52,0.7809,8,20,0
3,91.21,0.8485,1,0,0
4,216.42,0.4583,10,10,0
5,70.03,0.2019,40,19,1
6,223.6,0.6357,7,22,0
7,177.5,0.8531,2,6,0
8,57.69,0.773,6,15,0
9,75.77,0.1405,24,4,1
10,217.71,0.6118,10,2,0
11,103.23,0.5027,8,8,0
12,252.33,0.6957,5,21,0
13,220.52,0.4828,10,16,0
14,204.55,0.822,9,11,0
15,99.65,0.478,1,6,0
16,207.24,0.5393,0,3,0
17,267.04,0.6793,3,6,0
18,64.18,0.4462,35,3,1
19,111.54,0.7406,6,4,0
20,88.67,0.6343,6,16,0
```

2. Isolation Forest training & false-alarm identification

Method:

- Trained `sklearn.ensemble.IsolationForest` on numeric features only: `duration_sec`, `peak_amplitude`, `repetition_count`, `hour_of_day` (columns `id` and `is_false_alarm` excluded from training — unsupervised learning).
- `contamination=0.12` — prior approximate fraction of anomalies / nuisance alarms.
- **Outliers** (`predict == -1`) are treated as **anomalies**; for this assignment they are mapped in the API response to `likely_false_alarm` (homework narrative linking anomalies to suspicious / false-alarm-like behavior).
- Ground-truth column `is_false_alarm` is used **only** for optional offline comparison after training (not for fitting the forest).

Script: `train.py`

```
(.venv) jenniferha@Khues-Mac-Pro homework-3.2 % python train.py
Saved model to /Users/jenniferha/Desktop/Apply & Jobs/Antra/hw3-security-cloud-ai/homework-3.2/model/isolatio
n_forest.joblib
Saved feature metadata to /Users/jenniferha/Desktop/Apply & Jobs/Antra/hw3-security-cloud-ai/homework-3.2/mod
el/feature_columns.json

--- Illustrative: P(outlier flag) matches ground-truth is_false_alarm ---
Agreement rate: 0.906 (unsupervised model; not tuned for this label)
```

3. Model file on local disk

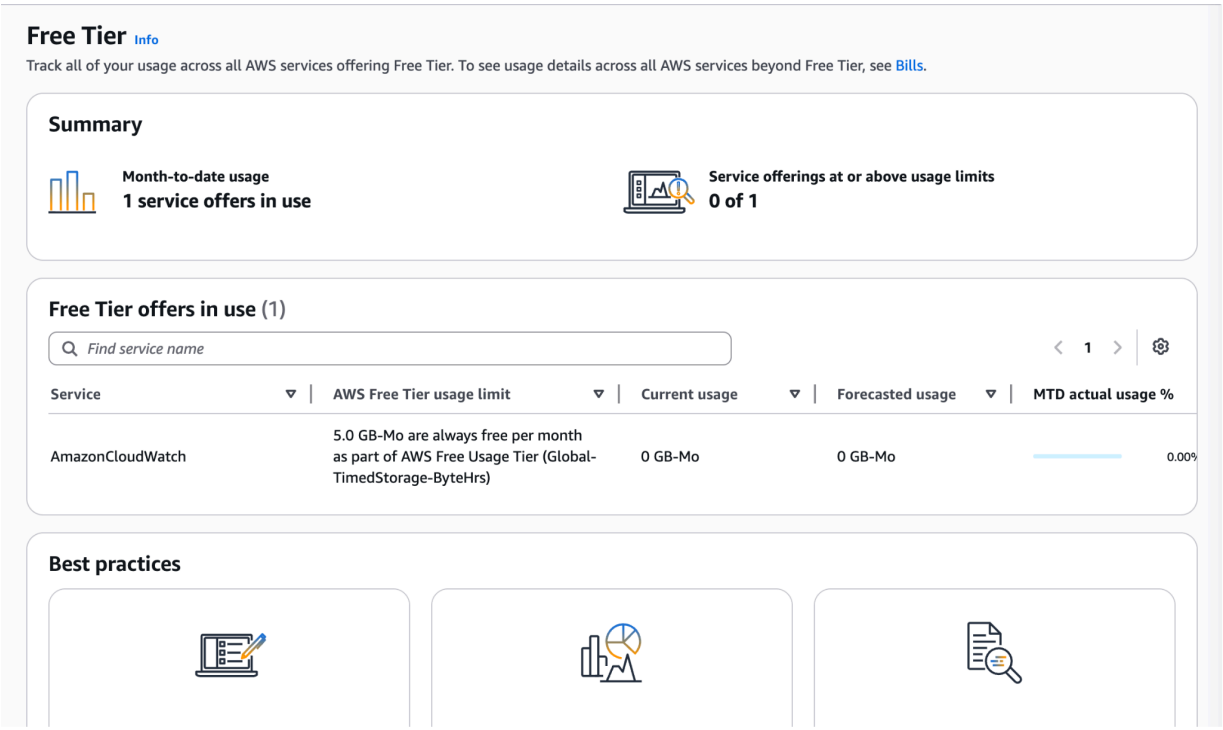
Artifacts:

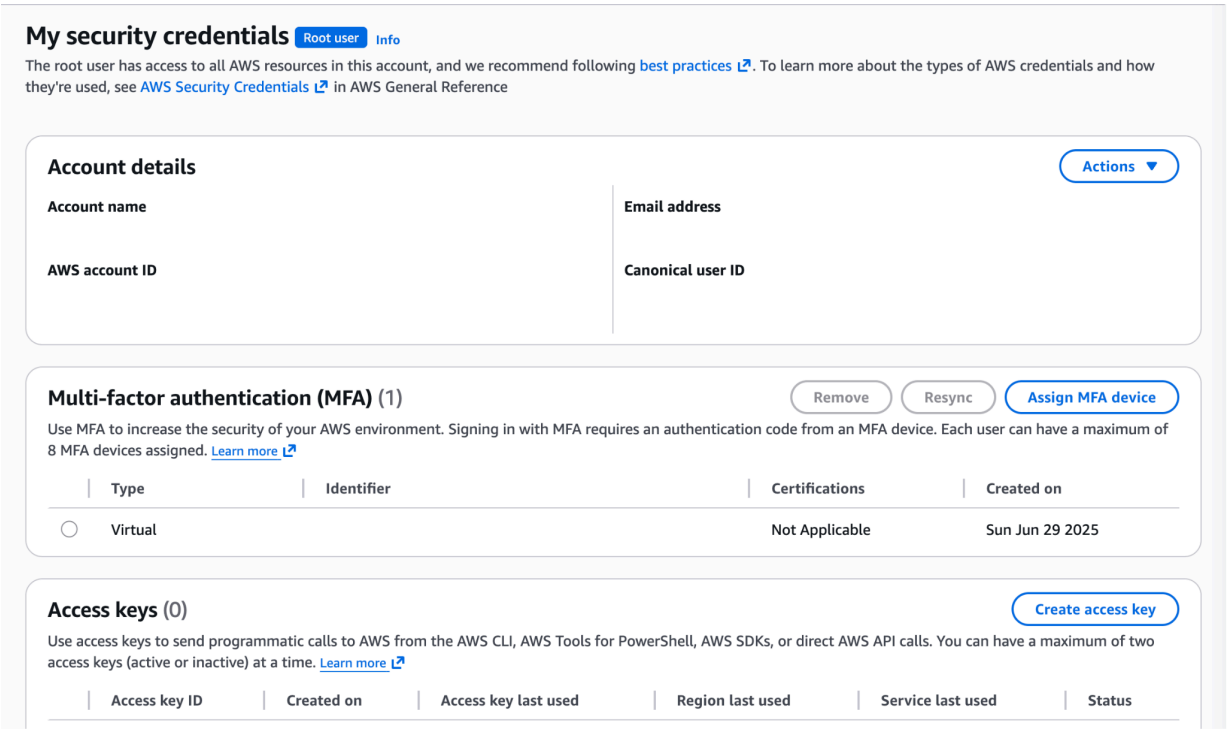
File	Description
`model/isolation_forest.joblib`	Serialized Isolation Forest (joblib).
`model/feature_columns.json`	Feature name order for inference in Lambda.

model	Today at 3:18 AM	--	Folder
feature_columns.json	Today at 4:55 AM	130 bytes	JSON
isolation_forest.joblib	Today at 4:55 AM	3.3 MB	Document

4. AWS Free Tier account

- Registered AWS account with **Free Tier** eligibility.
- Enabled **MFA** on the root or IAM admin user.
- Chose a primary **region** (e.g. `us-east-1`) for S3, Lambda, and HTTP API.

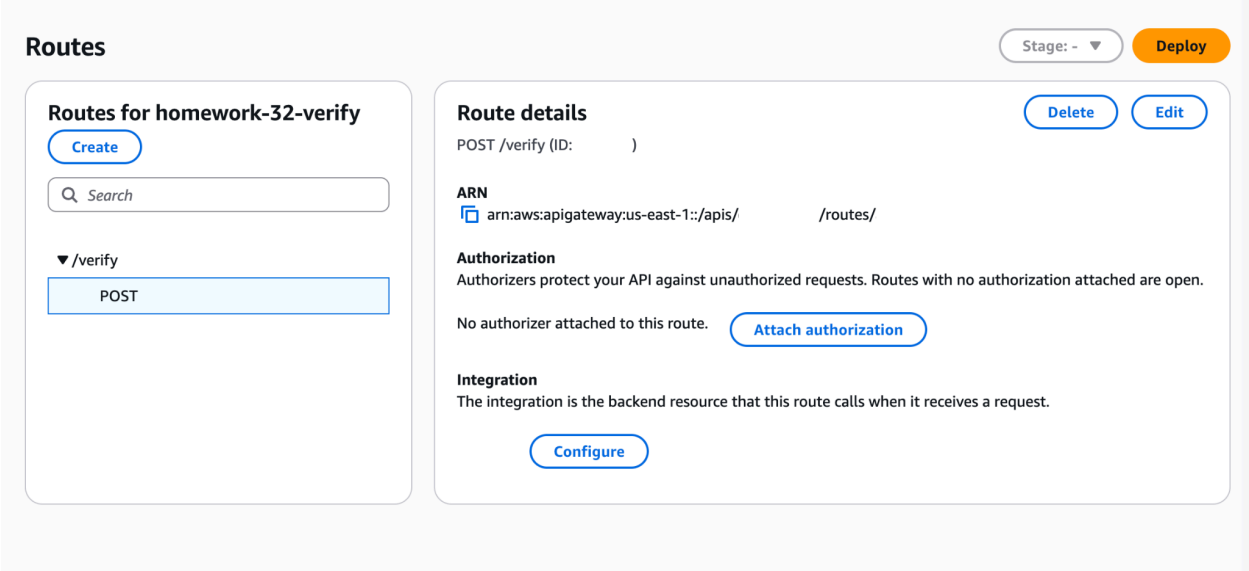




5. API Gateway — POST /verify with JSON body → Lambda

Implementation: AWS SAM template `template.yaml`

- **AWS::Serverless::Function** `VerifyFunction` with `HttpApi` event:
  - **Path:** `/verify`
  - **Method:** `POST`
- Implicit **HTTP API** (API Gateway v2) — output `HttpApiUrl` after deploy.



6. AWS SAM for local Lambda development

Commands used:

```
sam validate --region us-east-1 -t template.yaml
sam build --use-container
sam local start-api
```

Repository files: `template.yaml`, `events/verify-post.json`, `samconfig.toml.example`

```
● (.venv) jenniferha@Khues-Mac-Pro homework-3.2 % sam build --use-container
  Starting Build inside a container
  Cross-architecture build detected: Building x86_64 function 'VerifyFunction' on arm64 host. This may cause build failures with docker. Consider
  updating the function architecture to 'arm64' in template.yaml
  Building layer 'SklearnLayer'

  Fetching public.ecr.aws/sam/build-python3.12:latest-x86_64 Docker container image......
  Mounting /Users/jenniferha/Desktop/Apply & Jobs/Antra/hw3-security-cloud-ai/homework-3.2/layers/sklearn as /tmp/samcli/source:ro,delegated, inside
  runtime container
  SAM_CONTAINER_ID: a2288e1b4993c501f3454f75c1994fa565be1a50920285b886b2b17fccf7bc2a
   Running PythonPipBuilder:ResolveDependencies
   Running PythonPipBuilder:CopySource
  Building codeuri: /Users/jenniferha/Desktop/Apply & Jobs/Antra/hw3-security-cloud-ai/homework-3.2/lambda runtime: python3.12 architecture: x86_64
  functions: VerifyFunction

  Fetching public.ecr.aws/sam/build-python3.12:latest-x86_64 Docker container image......
  Mounting /Users/jenniferha/Desktop/Apply & Jobs/Antra/hw3-security-cloud-ai/homework-3.2/lambda as /tmp/samcli/source:ro,delegated, inside runtime
  container
  SAM_CONTAINER_ID: 1e344aeeadf5c4f61bdc131707caff46fb5227cdcc53c458975c1c1d713ef095

  Build Succeeded

  Built Artifacts  : .aws-sam/build
  Built Template   : .aws-sam/build/template.yaml

  Commands you can use next
  =========================
  [*] Validate SAM template: sam validate
  [*] Invoke Function: sam local invoke
  [*] Test Function in the Cloud: sam sync --stack-name {{stack-name}} --watch
  [*] Deploy: sam deploy --guided
   Running PythonPipBuilder:ResolveDependencies
   Running PythonPipBuilder:CopySource
```

```
● (.venv) jenniferha@Khues-Mac-Pro homework-3.2 % sam local start-api --env-vars env.json
  Mounting VerifyFunction at http://127.0.0.1:3000/verify [POST]
  You can now browse to the above endpoints to invoke your functions. You do not need to restart/reload SAM CLI while working on your functions,
  changes will be reflected instantly/automatically. If you used sam build before running local commands, you will need to re-run sam build for the
  changes to be picked up. You only need to restart SAM CLI if you update your AWS SAM template
  2026-05-14 08:01:59 WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
   * Running on http://127.0.0.1:3000
  2026-05-14 08:01:59 Press CTRL+C to quit
  Invoking app.lambda_handler (python3.12)
  SklearnLayer is a local Layer in the template
  Local image is up-to-date
  Building image.....................
  Using local image: samcli/lambda-python:3.12-x86_64-cffe7371f785532f399455d71.

  Mounting /Users/jenniferha/Desktop/Apply & Jobs/Antra/hw3-security-cloud-ai/homework-3.2/.aws-sam/build/VerifyFunction as /var/task:ro,delegated,
  inside runtime container
  SAM_CONTAINER_ID: e48da0fe816c173588ffa149103bd5826698cd989d7512c22f4acc8e9ac4c2b3
  START RequestId: a196b419-60e9-4c40-ba3b-2d01d8311857 Version: $LATEST
  END RequestId: a196b419-60e9-4c40-ba3b-2d01d8311857
  REPORT RequestId: a196b419-60e9-4c40-ba3b-2d01d8311857  Init Duration: 1.44 ms  Duration: 23664.71 ms   Billed Duration: 23665 ms       Memory Siz
  e: 1024 MB      Max Memory Used: 1024 MB

  2026-05-14 08:02:33 127.0.0.1 - - [14/May/2026 08:02:33] "POST /verify HTTP/1.1" 200 -
```

7. Upload model to Amazon S3

Procedure (after `sam deploy`):

1. Read stack output `ModelBucketName`.
2. Upload artifacts:

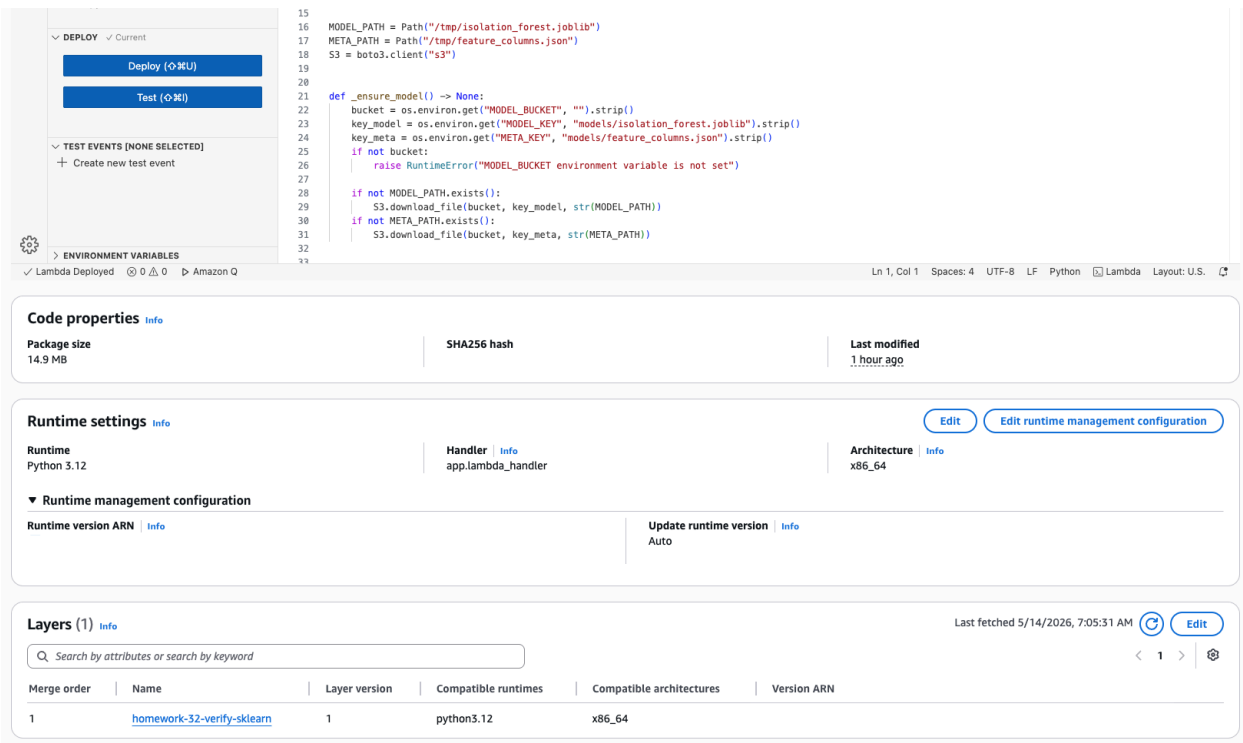
```
aws s3 cp model/isolation_forest.joblib
s3://<bucket>/models/isolation_forest.joblib

aws s3 cp model/feature_columns.json
s3://<bucket>/models/feature_columns.json
```

```
● (.venv) jenniferha@Khues-Mac-Pro homework-3.2 % cd "/Users/jenniferha/Desktop/Apply & Jobs/Antra/hw3-security-cloud-ai/homework-3.2"

  aws s3 cp model/isolation_forest.joblib s3://homework-32-verify-modelbucket-[redacted]/models/isolation_forest.joblib

  aws s3 cp model/feature_columns.json s3://homework-32-verify-modelbucket-[redacted]/models/feature_columns.json

  aws s3 ls s3://homework-32-verify-modelbucket-[redacted]/models/
  upload: model/isolation_forest.joblib to s3://homework-32-verify-modelbucket-[redacted]/models/isolation_forest.joblib
  upload: model/feature_columns.json to s3://homework-32-verify-modelbucket-[redacted]/models/feature_columns.json
  2026-05-14 07:39:28        130 feature_columns.json
  2026-05-14 07:39:27    3331433 isolation_forest.joblib
```

8. Lambda layer for libraries

Implementation:

- Directory `layers/sklearn/requirements.txt` lists: `numpy`, `scipy`, `scikit-learn`, `joblib` (pinned ranges compatible with Lambda Python 3.12).
- SAM resource `SklearnLayer` (`AWS::Serverless::LayerVersion`) with `Metadata: BuildMethod: python3.12` so `sam build` installs dependencies into the layer artifact.
- Function `VerifyFunction` references the layer via `Layers: [!Ref SklearnLayer]`.
- Function code `lambda/requirements.txt` keeps `boto3` for S3 download (runtime also includes boto3, but explicit is fine).



9. Lambda loads model from S3 and serves via API Gateway

Implementation (`lambda/app.py`):

1. Read environment variables `MODEL_BUCKET`, `MODEL_KEY`, `META_KEY` (set by SAM from template defaults + bucket ref).

2. On cold start, `boto3` `download_file` to `/tmp/isolation_forest.joblib` and `/tmp/feature_columns.json` (cached for warm invocations).
3. Parse JSON body; build the feature vector in `feature_columns.json` order.
4. Return JSON with `likely_false_alarm`, `isolation_forest_prediction`, `anomaly_score`, and a short interpretation string.

IAM: SAM policy `S3ReadPolicy` scoped to the stack's `ModelBucket`.

10. Test API with Postman

Steps:

1. **POST** `{{HttpApiUrl}}/verify`
2. Headers: `Content-Type: application/json`
3. Body (raw JSON), example:

```
{
  "duration_sec": 15.5,
  "peak_amplitude": 0.22,
  "repetition_count": 35,
  "hour_of_day": 3
}
```

4. Expect **HTTP 200** and a JSON body including `likely_false_alarm`: `true` or `false`.

https://euqdyt9r45.execute-api.us-east-1.amazonaws.com/verify

Save | Share

POST https:// .execute-api.us-east-1.amazonaws.com/verify Send

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

raw JSON Schema Beautify

```
1 {
2   "duration_sec": 15.5,
3   "peak_amplitude": 0.22,
4   "repetition_count": 35,
5   "hour_of_day": 3
6 }
```

Body 200 OK 10.65 s 375 B

{} JSON Preview Visualize

```
1 {
2   "likely_false_alarm": true,
3   "isolation_forest_prediction": -1,
4   "anomaly_score": -0.05726076289316118,
5   "interpretation": "prediction -1 means outlier (anomaly); mapped to
      likely_false_alarm for assignment narrative."
6 }
```

http://127.0.0.1:3000/verify

Save | Share

POST http://127.0.0.1:3000/verify Send

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

raw JSON Schema Beautify

```
1 {
2   "duration_sec": 15.5,
3   "peak_amplitude": 0.22,
4   "repetition_count": 35,
5   "hour_of_day": 3
6 }
```

Body 200 OK 29.37 s 384 B

{} JSON Preview Visualize

```
1 {
2   "likely_false_alarm": true,
3   "isolation_forest_prediction": -1,
4   "anomaly_score": -0.05726076289316118,
5   "interpretation": "prediction -1 means outlier (anomaly); mapped to
      likely_false_alarm for assignment narrative."
6 }
```